

Topological Foundations: Jordan Curve Theorem

Formalization Notes

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Definition 1 (Homotopy and Nullhomotopy). *If f and f' are continuous maps of the space X into the space Y , we say that f is **homotopic** to f' if there is a continuous map $F : X \times I \rightarrow Y$ such that*

$$F(x, 0) = f(x) \quad \text{and} \quad F(x, 1) = f'(x)$$

*for each $x \in X$. (Here $I = [0, 1]$.) The map F is called a **homotopy** between f and f' .*

*If f is homotopic to f' , we write $f \simeq f'$. If $f \simeq f'$ and f' is a constant map, we say that f is **nullhomotopic**.*

Definition 2 (Path Homotopy). *Two paths f and f' mapping the interval $I = [0, 1]$ into X are said to be **path homotopic** if they have the same initial point x_0 and the same final point x_1 , and if there is a continuous map $F : I \times I \rightarrow X$ such that:*

$$\begin{aligned} F(s, 0) &= f(s) \quad \text{and} \quad F(s, 1) = f'(s) \\ F(0, t) &= x_0 \quad \text{and} \quad F(1, t) = x_1 \end{aligned}$$

*for each $s \in I$ and each $t \in I$. We call F a **path homotopy** between f and f' . If f is path homotopic to f' , we write $f \simeq_p f'$.*

[Lemma 51.1] The relations $\simeq$ and $\simeq_p$ are equivalence relations. If f is a path, we shall denote its path-homotopy equivalence class by $[f]$.

Proof. Let us verify the properties of an equivalence relation.

Given f , it is trivial that $f \simeq f$; the map $F(x, t) = f(x)$ is the required homotopy. If f is a path, F is a path homotopy.

Given $f \simeq f'$, we show that $f' \simeq f$. Let F be a homotopy between f and f' . Then

$$G(x, t) = F(x, 1 - t)$$

is a homotopy between f' and f . If F is a path homotopy, so is G .

Suppose that $f \simeq f'$ and $f' \simeq f''$. We show that $f \simeq f''$. Let F be a homotopy between f and f' , and let F' be a homotopy between f' and f'' . Define $G : X \times I \rightarrow Y$ by the equation

$$G(x, t) = \begin{cases} F(x, 2t) & \text{for } t \in [0, \frac{1}{2}], \\ F'(x, 2t - 1) & \text{for } t \in [\frac{1}{2}, 1]. \end{cases}$$

The map G is well defined, since if $t = \frac{1}{2}$, we have $F(x, 2t) = f'(x) = F'(x, 2t - 1)$. Because G is continuous on the two closed subsets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$ of $X \times I$, it is continuous on all of $X \times I$, by the pasting lemma. Thus G is the required homotopy between f and f'' . The proof for $\simeq_p$ is entirely similar. $\square$